

A 60-GHz 19.8-mW Current-Reuse Active Phase Shifter With Tunable Current-Splitting Technique in 90-nm CMOS

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Abstract—A 60-GHz 4-bit vector-summing phase shifter has been implemented in a 90-nm CMOS process. Based on a tunable current-splitting technique with π -type low-pass filter (LPF) and T-type high-pass filter (HPF), a quadrature adjustable amplitude generator is developed to improve the insertion loss and adjust quadrature signals' amplitudes. Variable gain amplifiers (VGAs) that are widely used in traditional phase shifters are replaced by the generator to reduce power consumption. In addition, a balun-based current-reuse technique is applied to further save dc power. The overall consumed direct current is 11 mA under 1.8-V voltage supply. The measured results show that the phase shifter with an output balun (with loss of 1.6 dB) achieves gain of $-0.4\sim 2.5$ dB at 60 GHz and root mean square (rms) gain error is $0.75\sim 1.6$ dB in the frequency regime from 57 to 64 GHz. The peak average gain is 1.1 dB at 59.7 GHz. By employing a calibration method, the measured rms phase error ranges from 2.3° to 7.6° from 57 to 64 GHz. The tested 60-GHz input 1-dB compression point (P_{1dB}) and noise figure (NF) for 16 phase states are -9.8 ± 0.8 dBm and 11 ± 1.3 dB, respectively.

Index Terms—Active phase shifter, balun-based current-reuse technique, CMOS, current-splitting technique, millimeter-wave (mm-wave), RF phase shifting, RF integrated circuit (RFIC), 60 GHz, vector synthesizer.

I. INTRODUCTION

WITH the rapid development of silicon semiconductor technology and integrated circuit (IC) design methodologies, silicon-based millimeter wireless communication systems have made considerable progress in recent years [1]–[3]. However, the propagation distance of a 60-GHz wireless communication system is limited

by the high attenuation of millimeter-wave (mm-wave) electromagnetic (EM) waves due to strong oxygen absorption [2]. The phased array, which has advantages of enhancement of the signal-to-noise ratio (SNR) and effective isotropic radiated power (EIRP), becomes an attractive technique to improve the performance of high data-rate long/moderate-distance wireless communication and radar systems [4]–[19]. Several phased-array architectures are developed, including local oscillator (LO) phase shifting [5], [6], IF/baseband (BB) phase shifting [7]–[9], and RF phase shifting [10]–[15]. Since RF phase-shifting phased arrays have excellent feasibility for large-scale elements and possess high pattern directivity, it is quite prevalent in mm-wave CMOS phased-array design [12]–[15].

Phase shifters are key building blocks in phased-array transceivers. However, CMOS RF phase shifters face many technical challenges, such as high insertion loss in passive phase shifters and high power consumption in active ones. Passive phase shifters are usually implemented in two types, switched network [10], [14]–[16] and reflective type [12], [13], [17]–[19]. They possess good linearity without any power consumption. However, they suffer from large insertion loss at high frequencies and usually consume a big chip area. In [14] and [16], variable gain amplifiers (VGAs) are adopted to compensate the large insertion loss of passive phase shifters at the cost of degraded linearity and increased power consumption. Active phase shifters, which are based on the quadrature vector-summing technique, are also proposed for mm-wave phased-array applications [20]–[24]. The performance of quadrature signal generations has a crucial impact on phase-shifting accuracy of vector-summing phase shifters. In [22] and [25], RC poly-phase filters are created to generate a quadrature signal, but they encounter large insertion loss. Thus, they are not suitable to RF phase shifters, especially in the high-frequency regime. Quadrature all-pass filters (QAFs) [21], [23] and an inductive load quadrature filter [24] are proposed for mm-wave applications. However, these structures use resistors to get 90° phase difference between generated two-path signals and their insertion loss is also large in the high frequency band. As a result, active phase shifters consume high power consumption with uncompetitive gain.

Manuscript received August 19, 2015; revised January 26, 2016 and February 25, 2016; accepted March 9, 2016. Date of publication April 6, 2016; date of current version May 10, 2016. This work was supported by the National Science Fund of China under Grant 61422104, Grant 61331006, and Grant 61201004, and by the National Science and Technology Major Project of the Ministry of Science and Technology of China under Grant 2013ZX03001011. (Corresponding author: Kai Kang.)

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Digital Object Identifier 10.1109/TMTT.2016.2544306

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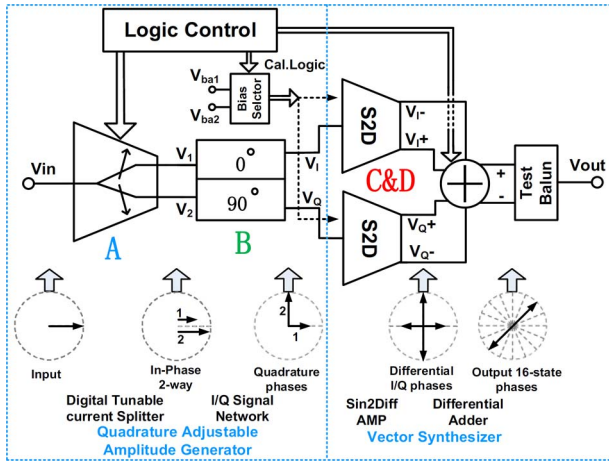


Fig. 1. Proposed architecture of mm-wave vector-sum phase shifter with an output balun.

In order to alleviate the above problems, a new architecture of an active phase shifter is proposed in this paper, as shown in Fig. 1. A quadrature adjustable amplitude generator, which consists of a tunable current splitter and an in-phase/quadrature-phase (I/Q) signal network is developed to effectively reduce insertion loss of quadrature signal generation and adjust the signals' amplitudes. It helps to eliminate the need for VGAs and digital-to-analog convertors (DACs). Compared with a conventional vector-summing phase shifter, the design complexity of the proposed one is also reduced. In addition, a balun based current-reuse technique is applied in the vector synthesizer, which is comprised of two single-to-differential amplifiers and a differential adder to reduce power consumption. Our work shows the on-chip implementation of an active phase shifter with good gain and low power consumption in the 60-GHz frequency regime [20].

In Section II, the fundamental principle of the proposed phase shifter is analyzed, including its phase error and gain error. Section III describes the detailed circuit implement of all blocks in this architecture. Measurement results and discussion of the phase shifter are presented in Section IV. Section V concludes this paper.

II. PRINCIPLE AND ANALYSIS OF ACTIVE mm-WAVE PHASE SHIFTER

The proposed active phase shifter is shown in Fig. 1, including a quadrature adjustable amplitude generator and a vector synthesizer. The generator consisting of a tunable current splitter and an I/Q signal network is used to create quadrature signals with adjustable amplitudes. The modulated I- and Q-path signals are then summed by a following vector synthesizer, which is constituted by two single-to-differential amplifiers and a differential adder.

A. Principle of the Proposed Active Phase Shifter

As illustrated in Fig. 1, the RF input signal V_{in} is split into two paths (V_1 & V_2) and the two path signals can be adjusted

by the tunable current splitter,

$$V_1 = \alpha(\vec{x}) V_{in} \quad (1)$$

$$V_2 = \beta(\vec{x}) V_{in} \quad (2)$$

where $\alpha(\vec{x})$ and $\beta(\vec{x})$, which are real, represent the variable gains of tunable current splitter's two paths. With logic controlling bits $\vec{x}$, the values of $\alpha(\vec{x})$ and $\beta(\vec{x})$ will be tuned. Through the I/Q signal network, 90° phase difference between the two paths is generated by

$$V_I = a e^{j\phi} V_1 = a e^{j\phi} \alpha(\vec{x}) V_{in} \quad (3)$$

$$V_Q = a e^{j(\phi+\pi/2)} V_2 = a e^{j(\phi+\pi/2)} \beta(\vec{x}) V_{in}. \quad (4)$$

The V_I , V_Q , a , and ϕ denote I-path voltage, Q-path voltage, insertion loss of I/Q signal network, and I-path phase delay, respectively. The quadrature signal generated so far is single ended and can only cover one quadrant. In order to reach fully 360° coverage, differential I/Q signals (V_I^+ , V_I^- , V_Q^+ , V_Q^-) are required. Therefore, two single-to-differential amplifiers are employed in the vector synthesizer. The amplifiers also amplify the differential I/Q signal with the same gain b in each path. The following differential adder sums the modulated I&Q paths and generates the required output signal phase, which is obtained by

$$V_{out}(\vec{x}) = a b e^{j(\phi+\pi x_i)} \alpha(\vec{x}) V_{in} + a b e^{j(\phi+\pi/2+\pi x_j)} \beta(\vec{x}) V_{in} \quad (5)$$

$x_i, x_j = 0 \text{ or } 1, \text{ and } \vec{x}$

where x_i and x_j are logic controlling bits for selecting polarities. The output signal's amplitude and phase are given by

$$|V_{out}| = a b V_{in} \sqrt{(\alpha(\vec{x}))^2 + (\beta(\vec{x}))^2} \quad (6)$$

$$\theta_{out} = \arctan\left(\frac{\beta(\vec{x})}{\alpha(\vec{x})}\right) + \frac{\pi}{2} |3x_j - x_i| + \phi. \quad (7)$$

Based on (7), the ratios between $\alpha(\vec{x})$ and $\beta(\vec{x})$ can be calculated for required phase resolution. For a 4-bit phase shifter, the values of $\alpha(\vec{x})/\beta(\vec{x}) = R_{Ideal}$ need to include 1/0, 0.71/0.29, 0.5/0.5, 0.29/0.71, and 0/1. They will guide the setting of the current tunable splitter's variable gain states.

B. Analysis of Phase Error

Actually, the relative amplitudes of the tunable current splitter's two output signals ($|V_1|/|V_2|$) will have deviation compared to theoretical value and its output signals have slightly accessional phase errors in some gain states. The I/Q signal network will also cause phase error and amplitude mismatch between the I and Q paths. To investigate their influence to overall performance, the two blocks' total quadrature phase error and amplitude mismatch between the I and Q paths are defined as $\Delta\theta$, $\Delta\eta$,¹ respectively. Substituting $\Delta\theta$ and $\Delta\eta$

¹ $20 \log_{10} \Delta\eta = 20 \log_{10}(V_{QR}/V_{IR}) - 20 \log_{10} R_{Ideal}$, where V_{QR} , V_{IR} denotes the I- and Q-path signal amplitudes with amplitude mismatch, respectively.

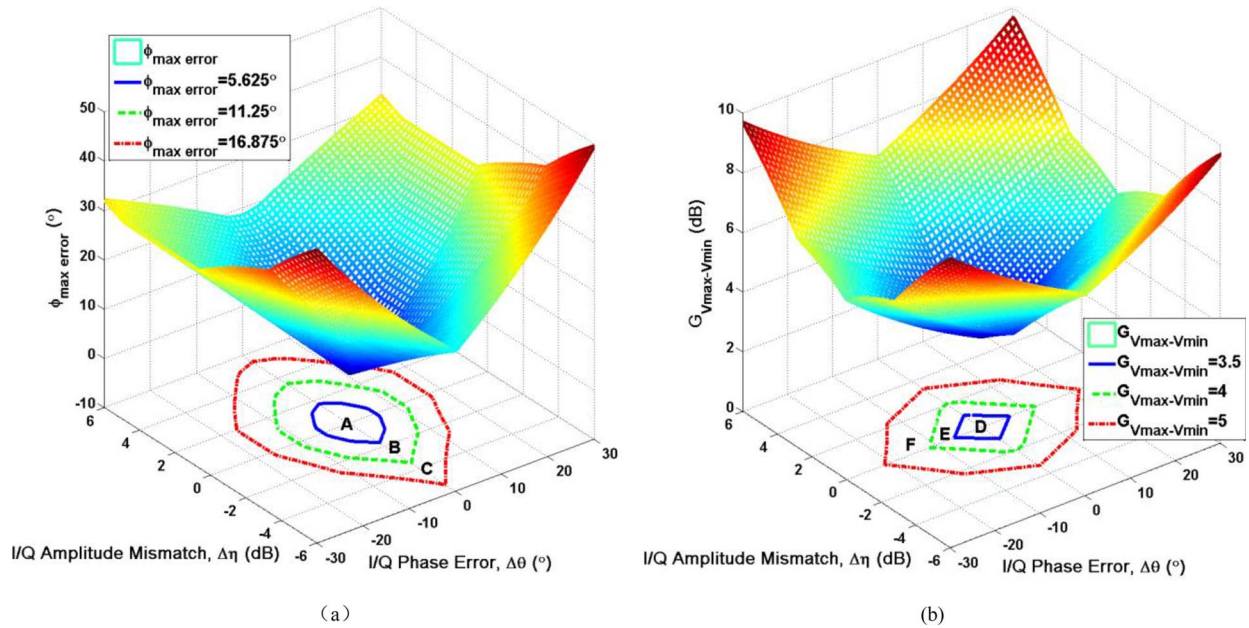


Fig. 2. (a) Maximum output phase error ($\phi_{\max \text{ error}}$) of the proposed phase shifter versus I/Q phase error ($\Delta\theta$) and I/Q total amplitude error ($\Delta\eta$); region A, B, and C are the places where $\phi_{\max \text{ error}} < 5.625^\circ$, $< 11.25^\circ$, and $< 16.875^\circ$, respectively. (b) Maximum gain error among all states ($G_{V \max} - V_{\min}$) versus $\Delta\theta$ and $\Delta\eta$; region D, E, and F are the areas where $G_{V \max} - V_{\min} < 3.5$ dB, < 4 dB, and < 5 dB, respectively.

into (5), the output signal is expressed as

$$V_{\text{out}} = abe^{j(\phi + \Delta\theta + \pi x_i)} \alpha(\vec{x}) V_{\text{in}} + \Delta\eta abe^{j(\phi + \pi/2 + \pi x_j)} \beta(\vec{x}) V_{\text{in}}. \quad (8)$$

The phase error is obtained by

$$\phi_{\text{error}}(\vec{x}) = \arctan\left(\frac{\beta(\vec{x})}{\alpha(\vec{x})}\right) - \arctan\left(\frac{(-1)^{(x_i - x_j)} \alpha(\vec{x}) \sin(\Delta\theta) + \Delta\eta \beta(\vec{x})}{\alpha(\vec{x}) \cos(\Delta\theta)}\right). \quad (9)$$

According to (9), the accuracy of the output-signal phase is strongly correlated to the precision of orthogonal phase and gain. As discussed in [23], for N -bit phase resolution, maximum output phase error ($\phi_{\max \text{ error}}$) is required to satisfy

$$\phi_{\max \text{ error}} < \frac{2\pi}{2^{N+1}} \\ \phi_{\max \text{ error}} = \max \left\{ \phi_{\text{error}}(\vec{x}) \mid \vec{x} \in \text{all states} \right\} (^\circ). \quad (10)$$

The trend of maximum output phase error of the 4-bit vector-summing phase shifter with $\Delta\theta$ and $\Delta\eta$ is presented in Fig. 2(a). The mesh and contours of $\phi_{\max \text{ error}}$ in Fig. 2(a) indicate that the margin of $\Delta\theta$ and $\Delta\eta$ will decrease with the growing requirement of phase resolution and precision. To the 4-bit vector-sum phase shifter, $\phi_{\max \text{ error}}$ needs to be smaller than 11.25° . Thus, the I/Q total amplitude error and I/Q phase error should locate in region B of Fig. 2(a), where $\Delta\theta < 10^\circ$ and $\Delta\eta < 3$ dB. In fact, making the I/Q amplitude mismatch in a certain range is more difficult than the quadrature phase error because all blocks will contribute to I/Q-path amplitude

mismatch. Therefore, a method by fine adjusting gain of single-to-differential amplifiers in the I&Q paths is used to compensate relative I/Q-path amplitude mismatch, which will be introduced in Section IV.

C. Analysis of Gain Error

In this architecture, the total voltage gain (G_0) of the tunable current splitter is almost equal for all gain states. Thus the sum of $\alpha(\vec{x})$ and $\beta(\vec{x})$ is assumed as constant. To obtain output signal phases of 0° , 22.5° , 45° , and 67.5° , taking the first quadrant as an instance, the splitter's two path voltage gains $\alpha(\vec{x})/\beta(\vec{x})$ needs to be set as $G_0/0$, $0.71G_0/0.29G_0$, $0.5G_0/0.5G_0$, and $0.29G_0/0.71G_0$. Substituting them to (6), the corresponding output signal amplitudes are abG_0V_{in} , $0.767abG_0V_{\text{in}}$, $0.707abG_0V_{\text{in}}$, and $0.767abG_0V_{\text{in}}$. Taking $\Delta\theta$ and $\Delta\eta$ into consideration, the output signal's voltage amplitude is

$$|V_{\text{out}}(\vec{x})| = abV_{\text{in}} \left[(\alpha(\vec{x}) \cos(\Delta\theta))^2 + \left(\alpha(\vec{x}) \sin(\Delta\theta) + (-1)^{|x_i - x_j|} \Delta\eta \beta(\vec{x}) \right)^2 \right]^{1/2}. \quad (11)$$

From (11), it can be observed that $\Delta\theta$ and $\Delta\eta$ will increase the difference of the output signal's amplitude among different phase states. In order to analyze the phase shifter's gain error, the maximum gain error ($G_{V \max} - V_{\min}$) is defined as

$$G_{V \max} - V_{\min} = 20 \log_{10} \max \{ |V_{\text{out}}(\vec{x})| \mid \vec{x} \in \text{all states} \} - 20 \log_{10} \min \{ |V_{\text{out}}(\vec{x})| \mid \vec{x} \in \text{all states} \}. \quad (12)$$

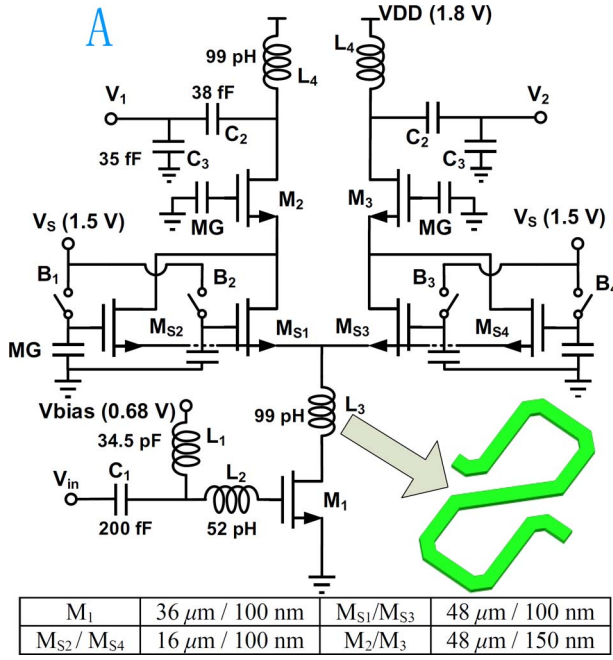


Fig. 3. Schematic of tunable current splitter and the layout of inserted inductor L_3 .

Fig. 2(b) shows the tendency of $G_{V \max} - V_{\min}$ with $\Delta\theta$ and $\Delta\eta$ increasing. The contours of $G_{V \max} - V_{\min}$ in Fig. 2(b) tell that achieving the maximum gain error ($G_{V \max} - V_{\min}$) less than 3.5, 4, and 5 dB, $\Delta\theta$ and $\Delta\eta$ need be inside region D, E, F, respectively. The theoretical analysis is compared with post-layout simulation results in Section III-D.

III. DESIGN OF THE mm-WAVE PHASE SHIFTER

As shown in Fig. 1, the proposed active phase shifter consists of a tunable current splitter, an I/Q signal network, two single-to-differential amplifiers, and a differential adder. The circuit with an output balun, which is used for single-ended measurement, is implemented in a 90-nm CMOS process.

A. Tunable Current Splitter

Fig. 3 shows the schematic of tunable current splitter in the quadrature adjustable amplitude generator. It is used to split the RF input signal into two in-phase branches and adjust their amplitudes. The splitter consists of a transconductance stage (M_1), a gain tuning stage ($M_{S1}-M_{S4}$), and an output stage (M_2-M_3). The passive inductors L_1 , L_2 , and capacitor C_1 are used to build up the input matching network. The splitter's gain is mainly dependent on the transconductance stage. Therefore, the size of transistor M_1 and its gate bias voltage is optimized through SPECTRE simulation. In order to enhance the circuit's gain, an inductor L_3 is inserted between M_1 and $M_{S1}-M_{S4}$ to resonate the transistors' parasitic capacitors including C_{ds} & C_{gd} of M_1 and C_{gs} & C_{ds} of $M_{S1}-M_{S4}$, which increases the gain by about 2 dB. In order to reduce chip area and improve the circuit symmetry, an S-shape inductor is designed (L_3 , as shown in the right bottom of Fig. 3). By turning on/off switches B_1-B_4 , the operating states of the

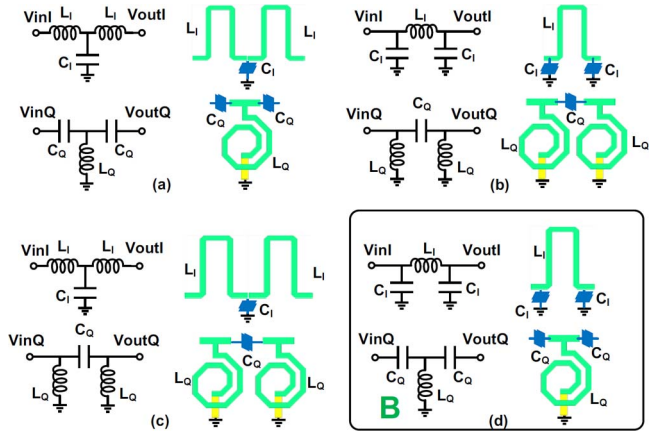


Fig. 4. Four combinations of three-element passive I/Q signal network and their simplified physical layout. (a) T-type high-pass filter (T-HPF) and T-type low-pass filter (T-LPF). (b) π -type low-pass filter (π -LPF) and π -type high-pass filter (π -HPF). (c) T-LPF and π -HPF. (d) π -LPF and T-HPF.

transistors $M_{S1}-M_{S4}$ can be controlled and the direct current in each branch are tuned. As a result, the two-path output RF signals with controllable amplitudes are generated. As discussed in Section II, for 4-bit phase resolution, the ratios of two paths' voltage gain $\alpha(\vec{x})/\beta(\vec{x})$ need be programmed as 1/0, 0.71/0.29, 0.5/0.5, 0.29/0.71, and 0/1. Thus the corresponding turning-on transistors are M_{S1} and M_{S2} , M_{S1} and M_{S4} , M_{S1} and M_{S3} , M_{S2} and M_{S3} , and M_{S3} and M_{S4} . To obtain 0.71/0.29 and 0.29/0.71 of $\alpha(\vec{x})/\beta(\vec{x})$, the sizes of M_{S1} (M_{S3}) and M_{S2} (M_{S4}) are chosen as 48 $\mu\text{m}/100$ nm (W/L) and 16 $\mu\text{m}/100$ nm with simulation optimization. The output impedances of the gain tuning stage are varying with different gain states. The variable output impedances increase phase errors and make output matching difficulty. Thus, the output stage (M_2 & M_3) is applied to alleviate these problems. The π -type network, which is made up of L_4 , C_2 , and C_3 is applied to match with the following stage. The simulation result shows that a single-path maximum gain of 2.5 dB is achieved with consuming 5-mA direct current under 1.8-V supply voltage.

B. I/Q Signal Network

Taking the advantages of this architecture, the I/Q signal network can be implemented by simple constant- k filters. As shown in Fig. 4, there are four combinations (T-HPF&T-LPF, π -LPF& π -HPF, T-LPF& π -HPF, and π -LPF&T-HPF) that can be applied to generate 90° phase difference between two signal paths. The LPF and HPF in either Fig. 4(c) or (d) have similar chip size because they have the same number of capacitors and inductors. Thus, compared to the former two networks [see Fig. 4(a) and (b)], they are easier to achieve symmetrical layout and same physical length from input to output terminals in I&Q paths. Symmetrical layout will effectively ensure the quadrature phase accuracy. The fourth network [see Fig. 4(d)] only uses two inductors. Therefore, the network that is comprised of π -LPF and T-HPF is more compact than the third one. The reason is that the inductors' physical structures are usually larger than the capacitors' in 60-GHz IC design. Due to these

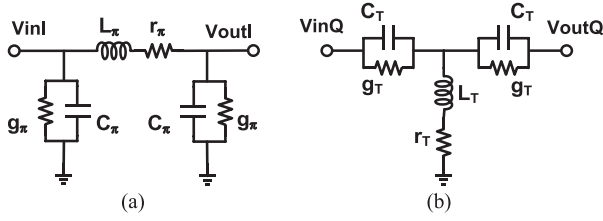


Fig. 5. Three-element networks with parasitic parameters. (a) π -LPF. (b) T-HPF.

advantages, the fourth topology [see Fig. 4(d)] is chosen to realize the I/Q signal network.

1) *Analysis of π -LPF and T-HPF*: The simplified models of π -LPF and T-HPF are presented in Fig. 5. The r_π , g_π , r_T , and g_T are used to represent the parasitic resistances/conductances of passive devices L_π , C_π , L_T , and C_T , respectively. The $ABCD$ matrices for the π -type and T-type networks (Fig. 5) are established as (13) and (14), shown at the bottom of this page [26], where Z_0 is characteristic impedance. Based on conversion formulas [27], the $ABCD$ matrices are converted to S matrices. The output impedance of the previous stage and the input impedance of the following stage need to be matched with Z_0 . Thus, $B = C$, which gives

$$(j\omega L_\pi + r_\pi) = \frac{2(j\omega C_\pi + g_\pi)Z_0^2}{1 - (j\omega C_\pi + g_\pi)^2 Z_0^2} \quad (15)$$

$$(j\omega L_T + r_T) = \frac{(j\omega C_T + g_T)^2 Z_0^2 - 1}{2(j\omega C_T + g_T)}. \quad (16)$$

Substituting $Q_{C\pi} = \omega C_\pi / g_\pi$ and $Q_{CT} = \omega C_T / g_T$ into (15) and (16), the phases of S_{21} of the two networks are calculated by

$$\begin{aligned} \phi_{\pi,T} &= \angle S_{21}^{\pi,T} \\ &= \arctan \left(\frac{2\omega C_{\pi,T} Z_0}{(\omega C_{\pi,T} Z_0)^2 \left(1 + \frac{1}{(Q_{C\pi,CT})^2} \right) - 1} \right). \end{aligned} \quad (17)$$

The two amplitudes of S_{21} are also obtained by

$$|S_{21}^{\pi,T}| = \frac{\sqrt{(2 - M_{\pi,T})^2 + 4(\omega C_{\pi,T} Z_0)^2}}{M_{\pi,T} + \left(\frac{2\omega C_{\pi,T} Z_0}{Q_{C\pi,CT}} \right)} \quad (18)$$

where $M_{\pi,T} = 1 + (\omega C_{\pi,T} Z_0)^2 (1 + 1/(Q_{C\pi,CT})^2)$.

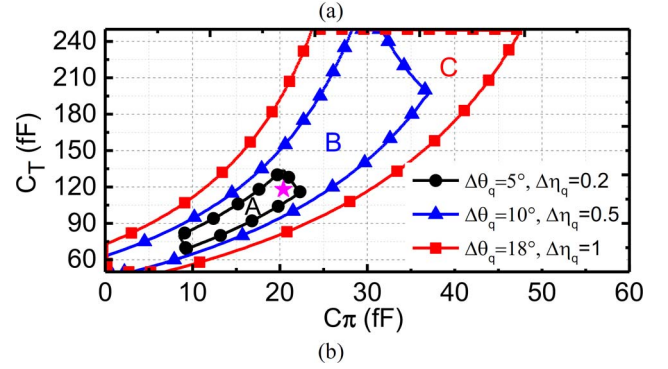
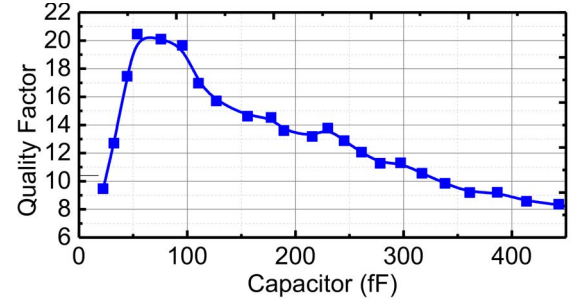


Fig. 6. (a) MiM capacitor' quality factor versus capacitance in 90-nm CMOS based on PDK model provided by the foundry. (b) Value ranges of C_T and C_π with quadrature phase error ($\Delta\theta_q$) and amplitude mismatch ($\Delta\eta_q$) for the 60-GHz I/Q signal network design. The star shows C_T and C_π used in this design ($Z_0 = 50 \Omega$, $Q_{\pi C} = 11$, $Q_{TC} = 16$).

Assume C_π , $Q_{C\pi}$, C_T , and Q_{CT} are defined, which will be fixed in (23) and (24). From (15), the inductor L_π in π -LPF is equal to

$$L_\pi = \frac{2M_\pi C_\pi Z_0^2}{M_\pi^2 - 4 \left(\frac{\omega C_\pi Z_0}{Q_{C\pi}} \right)^2}. \quad (19)$$

Meanwhile, its quality factor ($Q_{L\pi} = \omega L_\pi / r_\pi$) of L_π is calculated by

$$Q_{L\pi} = Q_{C\pi} \frac{M_\pi}{2 - M_\pi}. \quad (20)$$

Through (16), the inductance value of L_T in T-HPF can be acquired by

$$L_T = \frac{M_T}{2\omega^2 C_T \left(1 + \frac{1}{(Q_{CT})^2} \right)}. \quad (21)$$

The Q -factor of L_T , which is defined as $Q_{TL} = \omega L_T / r_T$, is thus given by

$$Q_{LT} = Q_{CT} \frac{M_T}{M_T - 2}. \quad (22)$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}_\pi = \begin{bmatrix} 1 + (j\omega L_\pi + r_\pi)(j\omega C_\pi + g_\pi) & \frac{(j\omega L_\pi + r_\pi)}{Z_0} \\ (2 + (j\omega L_\pi + r_\pi)(j\omega C_\pi + g_\pi))(j\omega C_\pi + g_\pi)Z_0 & 1 + (j\omega L_\pi + r_\pi)(j\omega C_\pi + g_\pi) \end{bmatrix} \quad (13)$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}_T = \begin{bmatrix} 1 + \frac{1}{(j\omega L_T + r_T)(j\omega C_T + g_T)} & \frac{1}{(j\omega C_T + g_T)Z_0} \left(2 + \frac{1}{(j\omega L_T + r_T)(j\omega C_T + g_T)} \right) \\ \frac{1}{(j\omega L_T + r_T)} & 1 + \frac{1}{(j\omega L_T + r_T)(j\omega C_T + g_T)} \end{bmatrix} \quad (14)$$

Since $Q_{L\pi}$ and Q_{LT} are positive, these two constrains $(\omega C_\pi Z_0)^2 < 1/(1 + 1/(Q_{C\pi})^2)$, $(\omega C_T Z_0)^2 > 1/(1 + 1/(Q_{CT})^2)$ should be satisfied in circuit design.

2) *I/Q Signal Network Design*: The ideal phase difference between π -LPF and T-HPF is $\pi/2$ or $3\pi/2$, and the amplitude response of π -LPF and T-HPF needs to be equal. Thus the two equations can be obtained

$$|\phi_T - \phi_\pi| = \frac{\pi}{2} \text{ or } \frac{3\pi}{2} \quad (23)$$

$$|S_{21}^\pi| = |S_{21}^T|. \quad (24)$$

From Fig. 6(a), the relationship between capacitances and their Q -factors can be obtained. Substituting (16) and (17) into (23) and (24), the capacitors C_π , C_T can be derived. Meanwhile, their Q -factors $Q_{C\pi}$, Q_{CT} are also solved [see Fig. 6(a)]. With C_π , C_T , $Q_{C\pi}$, and Q_{CT} , the inductors L_π , L_T , and their Q -factors $Q_{L\pi}$, Q_{LT} can be calculated from (19)–(22). In order to investigate the robustness of the I/Q signal network with the components value change, the quadrature phase error ($\Delta\theta_q$) and amplitude mismatch ($\Delta\eta_q$) of the network are defined as

$$\Delta\theta_q = |\phi_T - \phi_\pi| - \frac{\pi}{2} \text{ or } \frac{3\pi}{2} \quad (25)$$

$$\Delta\eta_q = |S_{21}^\pi| - |S_{21}^T|. \quad (26)$$

The contours of $\Delta\theta_q$ and $\Delta\eta_q$ with C_π and C_T are shown in Fig. 6(b) for a 60-GHz I/Q signal network. According to the contours in Fig. 6(b), the C_π , C_T should be located in region A, B, and C for $\Delta\theta_q < 5^\circ$ and $\Delta\eta_q < 0.2$ dB, $\Delta\theta_q < 10^\circ$ and $\Delta\eta_q < 0.5$ dB, and $\Delta\theta_q < 18^\circ$ and $\Delta\eta_q < 1$ dB, respectively. According to the analysis in Section II, for 4-bit phase resolution, the capacitances of C_π and C_T need be in region B with consideration of the amplitude error of the previous stage. From Fig. 6(b), we can also see the structure is strongly robust to the variation of components' values with acceptable phase and amplitude error.

Based on the above analysis, a wideband 60-GHz I/Q signal network is designed. The capacitor C_π is 20.4 fF with $Q_{C\pi}$ of 10.4 and the capacitor C_T is 118 fF with Q_{CT} of 15.8; the inductors L_π and L_T are 96 and 166.5 pH, and their Q -factors are 18.6 and 17, respectively. C_π and C_T are located in region A of Fig. 6. The chosen components' values ensure the network has relative large tolerability of process variations. According to the circuit simulation results (Fig. 7), from 54 to 68 GHz, the phase error is less than 5° and the amplitude error between the I- and Q-way is below 0.2 dB. The insertion loss of π -LPF and T-HPF are also low (< 0.8 dB). It guarantees the phase precision of the phase shifter.

C. Vector Synthesizer

Fig. 8(a) shows the schematic of the vector synthesizer. It consists of two single-to-differential amplifiers and a differential adder. Since the quadrature signal generated by the former stage is single-ended, before the adder, two single-to-differential amplifiers are used to transfer the single-ended I/Q-vector signal into differential mode (V_I^+ , V_I^- , V_Q^+ , V_Q^-) for fully reaching 360° coverage.

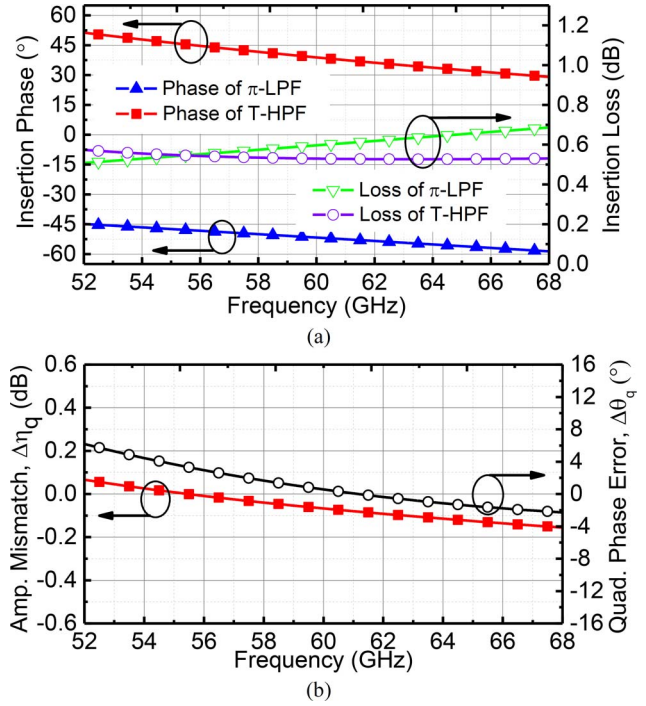


Fig. 7. (a) Simulated phase and insertion loss of π -LPF and T-HPF. (b) Simulated relative amplitude mismatch and quadrature phase error of the I/Q signal network.

They also amplify differential I/Q signals with the same value to further enhance the gain of the phase shifter. As shown in Fig. 8(b), the amplifier is designed by a common-source stage M_4 with a passive balun (Bn). The inductors L_5 , L_6 and source degeneration inductor L_7 compose the matching network with the former stage. The following differential adder is applied to sum the modulated I- and Q-path vector signal and synthesize the output signal phase. It consists of two polarity selectors and an output balun as common loads. The two selectors are composed of eight NMOS transistors [M_{A1} – M_{A4}] $\times$ 2 with inverters [top of Fig. 8(b)]. Through S1 and S2, the inverters switch on or off [M_{A1} – M_{A4}] $\times$ 2 by controlling the bias voltage of their gate terminals to select polarities of the I and Q signal to reach required quadrants.

To decrease power consumption, a balun based current-reuse technique is applied between the two single-to-differential amplifiers and the differential adder. As illustrated in Fig. 8(b), the direct current from sources of M_{A1} – M_{A4} enters the drain of M_4 through a balun (Bn). The balun's center tap of primary winding and one terminal of second winding are connected together to produce a path for direct current. At the balun's center tap, a group of on-chip capacitors whose total capacitance is 3.6 pF are used to create ac ground to RF signal.

Two top metals are utilized to design the balun (Bn) whose 3-D structure is shown in Fig. 8(c). Its inner diameter and line width of both windings are 57 and 5 μm , respectively. It is worth pointing out that the phase error and amplitude mismatch of the differential port are strongly correlated with the center tap's length. The full wave simulation using HFSS indicates that the balun in this work has less than 3° phase

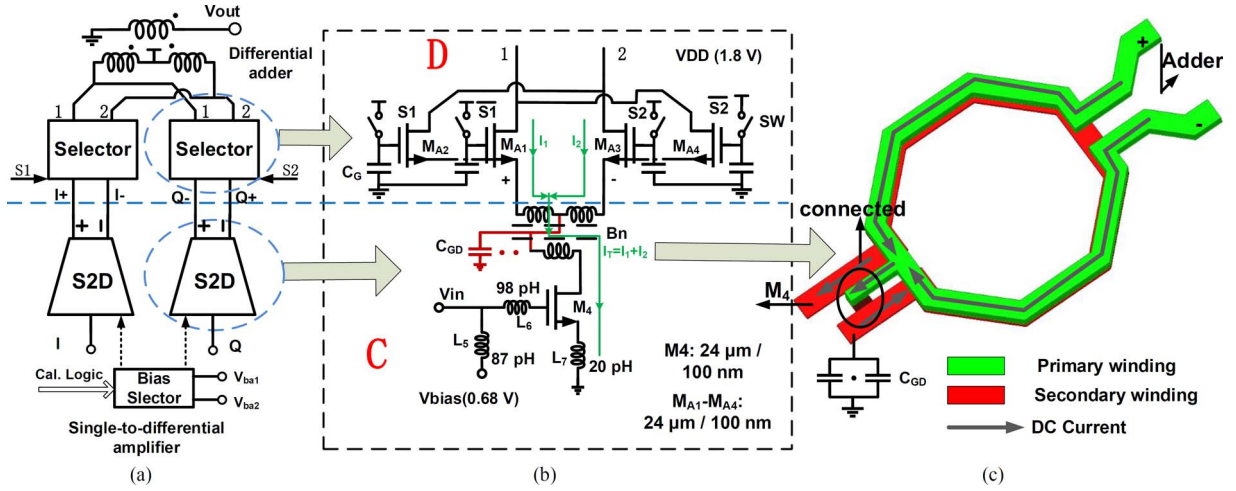


Fig. 8. (a) Vector synthesizer's structure, including two single-to-differential amplifiers and a differential adder. (b) Schematics of single-to-differential amplifier (bottom) and polarity selector (top). The green lines show the direct current's flowing direction. (c) 3-D view of current-reuse balun.

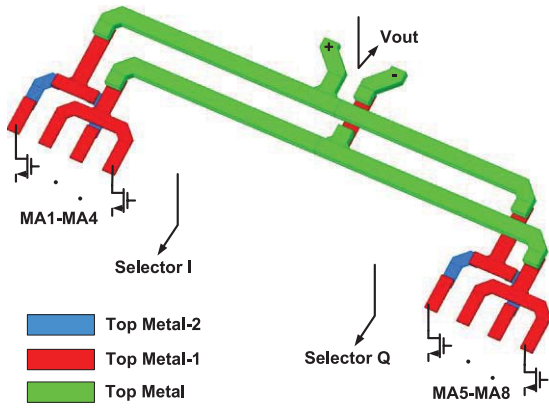


Fig. 9. Tower-like passive combination network of differential adder.

error and 0.2-dB amplitude mismatch with 10- μm tap length. Its quality factor is about 8.

As shown in Fig. 9, a *tower-like* passive combination network is design by three top metals, which is used to connect the eight transistors ($M_{A1}-M_{A4}$) $\times$ 2 of the two selectors. It provides a high consistency of time delay among each signal path and guarantees the balance of the circuit. It also has a compact physical size (50 μm $\times$ 200 μm).

Due to the limitation of the single-ended measurement instrument, an on-chip output balun (as shown in Fig. 10) is adopted to convert the differential output signal into a single-ended mode. It is comprised of a transformer and two short-circuit transmission lines [28]. The output balun is also a part of output matching network and has insertion loss of 1.6 dB estimated by EM simulation.

By employing the balun-based current-reuse technique, the total direct current of the two blocks is decreased to 6 mA. The simulated maximum gain is 0.8 dB at 61 GHz in each path.

D. Comparison Between Theoretical Analysis and Simulation

The maximum I/Q phase error ($\Delta\theta_{\max} = \max\{\Delta\theta(\vec{x}) | \vec{x} \in 5 \text{ states}\}$) and maximum I/Q amplitude mismatch ($\Delta\eta_{\max} = \max\{\Delta\eta(\vec{x}) | \vec{x} \in 5 \text{ states}\}$) among the five gain

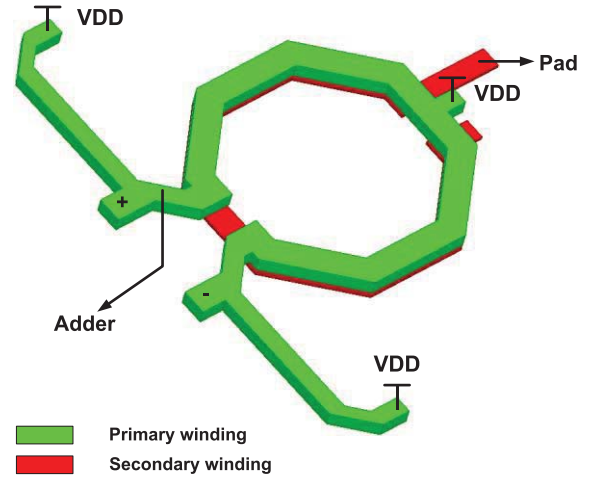


Fig. 10. 3-D structure of output balun. Its primary and secondary windings' line width and inner diameter are 6 and 51 μm , respectively. The length of the short-circuit transmission line is 47 μm with 4- μm line width.

states based on post-layout simulation, which are contributed by both the tunable current splitter and the I/Q signal network, are presented in Fig. 11(a). $\Delta\theta_{\max}$ is smaller than 6° and $\Delta\eta_{\max}$ is 1.9 ± 0.5 dB in the whole frequency band. The maximum I/Q amplitude mismatch happened when the tunable current splitter works on 0.71/0.29 and 0.29/0.71 gain states. This is mainly caused by the little difference between the two paths' output impedances of the splitter with M_{S1} & M_{S4} (or M_{S2} & M_{S3}) turning on. From 57 to 66 GHz, $\Delta\theta_{\max}$ and $\Delta\eta_{\max}$ are located in Region B of Fig. 2(a) and Region F of Fig. 2(b), where theoretical $\phi_{\max}$ error $< 11.25^\circ$ and $G_{V \max} - V_{\min} < 5$ dB, respectively. The simulated $\phi_{\max}$ error and $G_{V \max} - V_{\min}$, which are shown in Fig. 11(b), agree well with the theoretical values. Therefore, the theoretical analysis in Section II is efficient to estimate the performance of the phase shifter and is also useful to guide the phase-shifter design.

IV. MEASUREMENT RESULTS AND DISCUSSIONS

The proposed phase shifter has been implemented in commercial 90-nm CMOS technology. The fabricated die

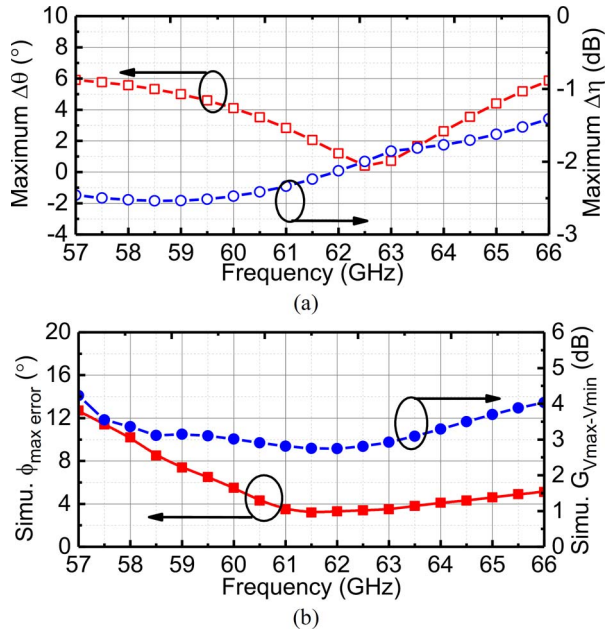


Fig. 11. (a) Simulated maximum I/Q phase error ($\Delta\theta_{\max}$) and maximum I/Q amplitude mismatch ($\Delta\eta_{\max}$). (b) Simulated maximum output phase error ($\phi_{\max}$ error) and maximum gain error ($G_{V_{\max}} - V_{\min}$).

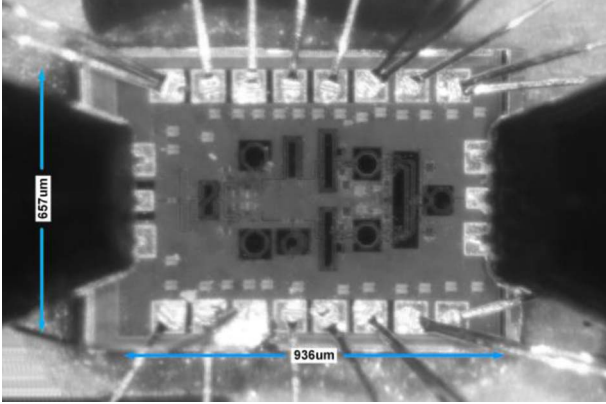


Fig. 12. Chip micrograph of 60-GHz active phase shifter.

micrograph is presented in Fig. 12. With all pads and the output balun, its size is $936 \mu\text{m} \times 657 \mu\text{m}$.

The S -parameter measurements for the phase shifter are done using an Agilent N5247A vector network analyzer. The dc and ground pads of the chips are bonded to a printed circuit board (PCB) (COB) (Fig. 12). A standard short-open-load-thru (SOLT) calibration to Cascade infinity microwave ground-signal-ground (G-S-G) probes tips is made using Cascade calibration substrate 101-190. A Niosecom NC5115 50-75-GHz noise source and an R&S FS67 spectrum analyzer with noise figure measurement function FS-K30 are used to test the noise figure of the phase shifter. A V-band 30-dB low-noise amplifier (LNA) as a pre-amplifier is used to provide enough gain to minimize measurement uncertainties of the whole NF measurement system. The classic Y -factor method is used to measure the NF of the phase shifter [29].

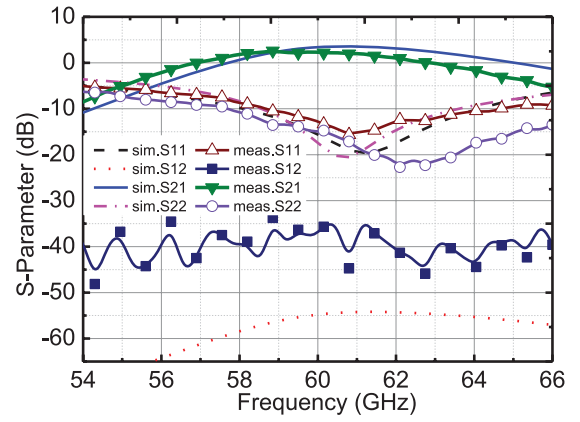


Fig. 13. Measured and simulated S -parameters of the phase shifter's reference (0°) state.

TABLE I
MEASURED PHASE STATES WITH LARGE PHASE ERROR AT 60 GHz

Phase State	Tunable Current Splitter Gain States				Phase Error without Cal.	Cal. (Y/N)	Phase Error with Cal.
	M_{S1}	M_{S2}	M_{S3}	M_{S4}			
67.5°	OFF	ON	ON	OFF	9.3°	Yes	4.5°
112.5°	OFF	ON	ON	OFF	6°	Yes	3.5°
202.5°	ON	OFF	OFF	ON	5.6°	Yes	2.4°
247.5°	OFF	ON	ON	OFF	9.1°	Yes	3.9°
270°	OFF	OFF	ON	ON	5.8°	No	N/A
315°	ON	OFF	ON	OFF	8.7°	Yes	0.6°
337.5°	ON	OFF	OFF	ON	8°	Yes	3.4°

For the 60-GHz phase shifter, Fig. 13 shows the measured S -parameters of reference (0°) state, the peak gain is 2.6 dB at 59.8 GHz with good input and output matching, and the 3-dB bandwidth is about 7 GHz from 56.5 to 63.5 GHz. The measurement results agree well with the simulations.

The 16 input matching and output matching characteristics for 4-bit phase states, which are controlled by off-chip digital logic bits, are presented in Fig. 14(a) and (b), respectively. S_{11} are smaller than -10 dB at 59–65 GHz and S_{22} are less than -10 dB at 58–66 GHz.

Fig. 15(a) shows the measurement results of 16 insertion phases. The measured maximum output phase error is 7° – 16° from 57 to 66 GHz and 10° at 60 GHz. From Table I, it can be observed that the majority of phase states with relative large phase error ($> 5^\circ$) are working on 0.71/0.29 or 0.29/0.71 gain state of the tunable current splitter. According to the discussion in Section III-D, the two gain states have relative big I/Q amplitude mismatch ($\Delta\eta$) and it leads to large phase error. In order to reduce phase error, a fix-tune off-chip calibration method by compensating I/Q amplitude mismatch in some phase states is applied (Table I). As shown in Fig. 8(a), the main idea is to fine adjust the g_m of M_4 in both the I and Q paths by each using two selectable gate bias voltages ($V_{\text{bal}} = 0.68$ V & $V_{\text{ba2}} = 0.72$ V) for M_4 . The bias voltages are directly offered by dc voltage source. According to the simulation results, increasing the gate bias voltage from 0.68 to 0.72 V will improve the single-to-differential

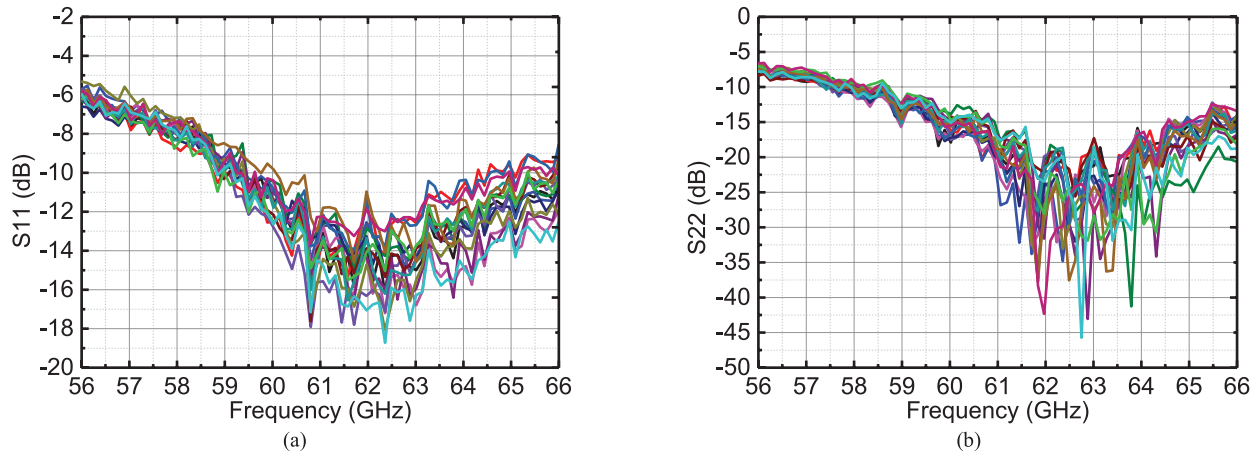


Fig. 14. Measured input return loss (S_{11}) and output return loss (S_{22}) of the phase shifter for 16 phase states. (a) S_{11} . (b) S_{22} .

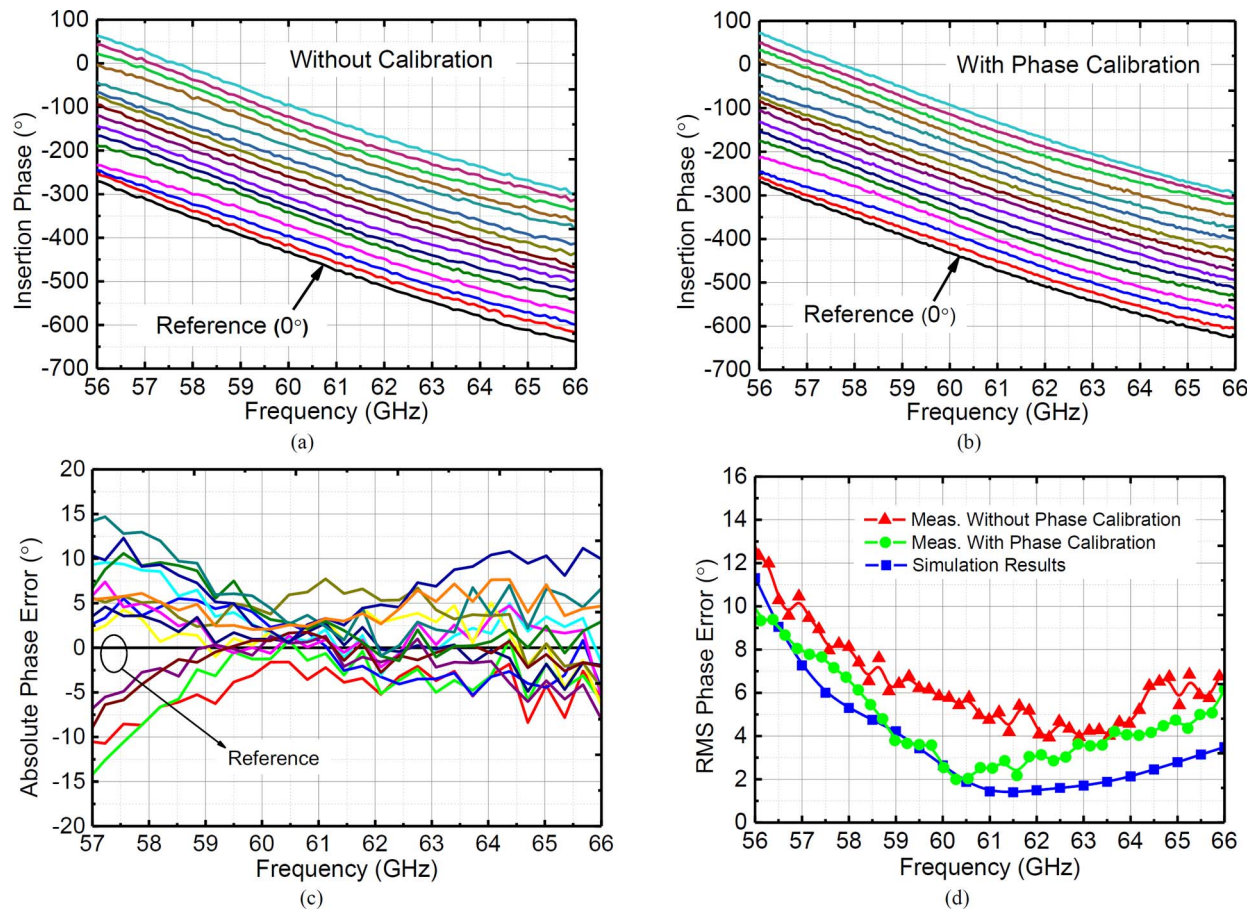


Fig. 15. Measured 16-state insertion phase and gain of the proposed phase shifter. (a) 16-state insertion phase without phase calibration. (b) 16-state insertion phase with phase calibration. (c) Absolute phase errors with the phase calibration versus phase states. (d) Comparison of rms phase errors of simulations, measurements without phase calibration, and measurements with phase calibration.

amplifier's gain by about 1.1 dB. Therefore, the I/Q amplitude mismatch will be reduced. The measured 16-state insertion phases and the absolute phase errors with the calibration are presented in Fig. 15(b) and (c), respectively. It is seen that the phase precision of the phase shifter is improved and the maximum phase error is reduced to 6° at 60 GHz. Fig. 15(d) shows the simulated, non-calibration, and calibrated measured

root mean square (rms) phase errors. The rms errors are calculated based on formulas defined in [21]. Without the calibration, the rms phase error variations are minimum 4.5° at 62.2 GHz and maximum 10° at 57 GHz. After applying the calibration, the rms phase error is from 2.3° to 7.6° in the entire frequency band. However, the measured rms phase errors are higher than the simulated one. The main reasons for that may

TABLE II
COMPARISON TABLE OF RF PHASE SHIFTERS

		JSSC 2007 [21]	TMTT 2013 [16]	JSSC 2013 [11]	RFIC 2009 [19]	TMTT 2012 [23]	JSSC 2012 [14]	This Work	
Technology		0.13- μm CMOS (1P8M)	90 nm CMOS	0.18- μm SiGe BiCMOS	0.13- μm SiGe BiCMOS	0.13- μm SiGe BiCMOS	0.13- μm SiGe BiCMOS	90 nm CMOS	
Method		Vector Summing	Switched Network+VGAs	Vector Summing	Vector Summing	Vector Summing+LNA	Switched Network+VGAs	Proposed Vector Summing	
Frequencies (GHz)		15-26	58-65	82-105	57-64	60-80	76-84	57-64	
Phase Range		360°	360°	360°	360°	360°	360°	360°	
Phase Resolution (bit)		4	5	4	5	4	4	4	
Peak Gain (avg.) (dB)	with amplifier	-3	-5.4	19**	-2	14.7	18.9	1.1	
	without amplifier	-3	-14.6	N/A	-2	-3.2***	-17***	1.1	
RMS Gain Error (dB)		1.1~2.1	<2	0.57 @ 90 GHz	N/A	1.3	0.6~1.1	0.75~1.6	
RMS Phase Error		6.5~13°	3.3~9.9°	0.27 @ 90 GHz	3°	<9.1°	3.9~10.4°	4.5~10°	2.3~7.6°
Phase Calibration		No	Yes	Yes	N/A	No	Yes	No	Yes
Noise Figure (dB)		N/A	N/A	<7 (RX)	~17	9~11.6	10.5±0.5 @ 80 GHz	9.7~13	
Input P _{1dB} (dBm)		-0.8	2	N/A	-10 @ 60 GHz	-27 @ 70 GHz	-26.7~23	-9.8±0.8 @ 60 GHz	
Input Return Loss (dB)		<-10 @ 16.8-26 GHz	N/A	N/A	N/A	<-10 @ 60-80 GHz	<-10 @ 70-88 GHz	<-10 @ 58.8-64.3 GHz	
Output Return Loss (dB)		<-10 @ 17-26 GHz	N/A	N/A	N/A	<-10 @ 60-73 GHz	<-10 @ 74-88 GHz	<-10 @ 57-66 GHz	
Power Consumption (mW)		11.7	31.2	N/A	32.4	108 (Phase shifter: 34.8)	32.4	19.8	
Chip Size (mm ²)		0.45	0.58*	N/A	0.56	1.06	1.28	0.61	

* without pads; **maximum gain; *** simulation result

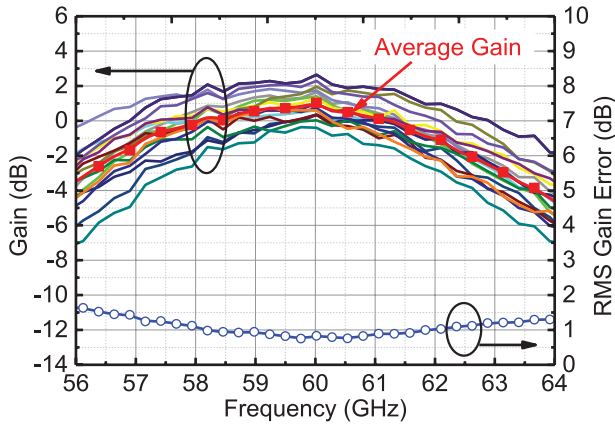


Fig. 16. Insertion power gain and rms gain error with the phase calibration.

be the deviation of EM simulation for passive structures and inadequate equivalent model for devices around 60 GHz.

The measured 16-state gain after the phase calibration from 56 to 64 GHz is shown in Fig. 16. The peak average gain of 1.1 dB is achieved at 59.7 GHz, including the on-chip output balun. As shown in Fig. 16, the measured rms gain error ranges from 0.75 dB at 59.8 GHz to 1.6 dB at 57 GHz. At 60 GHz, the gain with the output balun (with loss of 1.6 dB) is -0.4 – 2.5 dB for 16 phase states. The measured maximum gain error among all the phase states is 2.9 dB.

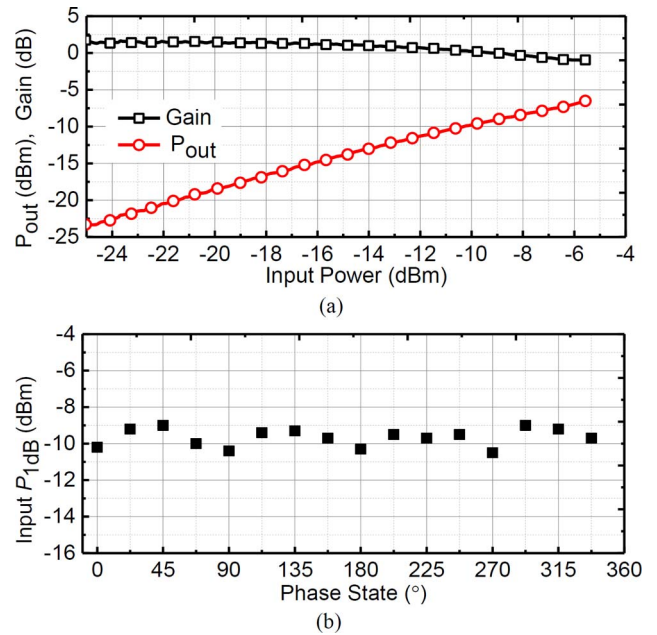


Fig. 17. (a) Nominal gain state's (45°) input $P_{1\text{ dB}}$ and power gain. (b) Measured input $P_{1\text{ dB}}$ of 16 phase states at 60 GHz.

Without using the phase calibration, the maximum gain error is 3.3 dB. After implementing the phase calibration, both phase error and gain error are improved simultaneously due to the

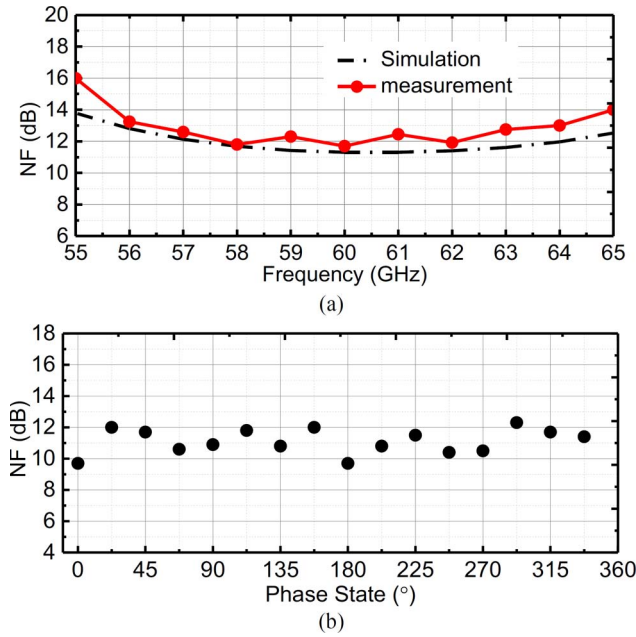


Fig. 18. (a) Nominal gain state's (45°) NF. (b) Measured NF of 4-bit phase states at 60 GHz.

decreasing of I/Q amplitude mismatch. It is consistent with the analysis in Section II. Therefore, the theoretical analysis is further validated by the measurement results.

The input P_1 dB and NF of the phase shifter are also tested. The nominal gain state's (45°) input P_1 dB is presented in Fig. 17(a). The measured input P_1 dB are from -10.6 to -9 dBm at 60 GHz depending on the phase states [see Fig. 17(b)]. Fig. 18(a) shows the nominal gain state's NF is < 13 dB from 56 to 65 GHz. At 60 GHz, the measured NF of 16 phase states shown in Fig. 18(b) is between 9.7 dB and 12.3 dB.

The chip performance is summarized and compared with the state-of-the-arts in Table II. The proposed 4-bit active phase shifter achieved high gain (peak average gain = 1.1 dB) with about 19.8-mW power consumption. The measured rms phase error is also less than other works.

V. CONCLUSION

This paper has presented a 60-GHz vector-summing phase shifter, which is demonstrated in 90-nm CMOS. Based on the tunable current-splitting technique with π -LPF and T-HPF, the quadrature adjustable amplitude generator is created to generate a quadrature signal with adjustable amplitude. It consists of the tunable current splitter and the I/Q signal network. Thanks to the low insertion-loss I/Q signal network, the generator achieves good gain and wide phase range. To reduce dc power consumption, the balun-based current-reuse technique is applied in the vector synthesizer. The whole circuit consumes only about 19.8 mW. The measured characteristics show that the proposed phase shifter achieves insertion gain of -0.4 – 2.5 dB at 60 GHz and rms gain error of 0.75 – 1.6 dB at 57–64 GHz. With the phase calibration method, phase errors are decreased and the rms phase error is from 2.3° to 7.6° over the entire frequency range. It is worth noting that

adding more tunable cells of the tunable current splitter will improve the phase resolution of the phase shifter. The excellent performance of the proposed active phase shifter indicates that it is suitable for integrated mm-wave phase-array systems.

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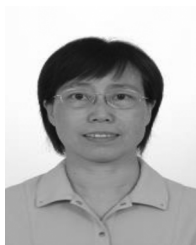
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